

1.2 Memory and storage

Why this unit is taught the way it is

What this unit teaches, and where it ends

18 lessons: 15 with an online experience, 2 paper assessments and 1 bonus challenge. 225 activities in all, worth 328 marks.

By the end of it a pupil should be able to:

- understand the need for primary storage ms-101
- know the difference between RAM and ROM ms-101
- know the purpose of RAM and ROM ms-101
- understand the need for virtual memory ms-102
- understand how virtual memory works ms-102
- understand the need for secondary storage ms-103
- know the common types of storage ms-103
- know how each type works ms-103
- know the characteristics of storage devices ms-104
- be able to compare devices using those characteristics ms-104
- understand the suitability of storage devices for certain applications ms-105
- understand the advantages and disadvantages of devices based on their characteristics ms-105
- know the units bit, nibble, byte, kilobyte, megabyte, gigabyte, terabyte and petabyte ms-106
- be able to convert between units ms-106
- understand that data must be converted to binary to be processed ms-106
- know what data capacity means ms-107
- understand how to calculate data capacity requirements ms-107
- convert positive denary whole numbers (0-255) to 8-bit binary and back ms-108
- add two 8-bit binary integers ms-108
- understand the terms most and least significant bit ms-108
- understand how overflow errors occur ms-108
- know how to perform left and right binary shifts ms-110
- understand what a binary shift achieves ms-110
- convert denary to 2-digit hexadecimal and back ms-110
- convert between binary and hexadecimal ms-110
- know which parts of the topic I am secure on ms-111
- finish any outstanding tasks from lessons 1 to 10 ms-111
- understand that all data, including text, is represented in binary ms-112
- know what is meant by a character set ms-112
- understand the relationship between the number of bits and the number of characters ms-112
- know about ASCII and Unicode ms-112
- understand how an image is a series of pixels represented in binary ms-113
- know what is meant by metadata ms-113
- understand the effect of colour depth and resolution on file size ms-113
- understand how sound is sampled and stored in digital form ms-114

- understand how sample rate, duration and bit depth affect file size and quality ms-114
- know why data is compressed ms-115
- understand the difference between lossy and lossless compression ms-115
- know why some data can only be compressed losslessly ms-115
- know which parts of data representation I am secure on ms-116
- finish outstanding tasks from lessons 12 to 15 ms-116

Why the lessons are in this order

Each lesson's starter retrieves the one before it, so the order is the retrieval schedule as much as the teaching one. The chain for this unit:

LESSON	TEACHES	RETRIEVES
1 Memory bay	understand the need for primary storage	nothing before it in this unit
2 Swap space	understand the need for virtual memory	Memory bay
3 Storage warehouse	understand the need for secondary storage	Swap space
4 Top Trumps	know the characteristics of storage devices	Storage warehouse
5 Pick the storage	understand the suitability of storage devices for certain applications	Top Trumps
6 Units lab	know the units bit, nibble, byte, kilobyte, megabyte, gigabyte, terabyte and petabyte	Pick the storage
7 Capacity calculator	know what data capacity means	Units lab
8 Binary workshop	convert positive denary whole numbers (0-255) to 8-bit binary and back	Capacity calculator
9 Assessment 1	assesses the whole unit	Binary workshop
10 Shift and hex lab	know how to perform left and right binary shifts	Assessment 1
11 Revision HQ 1.2 part 1	know which parts of the topic I am secure on	Shift and hex lab
12 Character foundry	understand that all data, including text, is represented in binary	Revision HQ 1.2 part 1
13 Pixel studio	understand how an image is a series of pixels represented in binary	Character foundry
14 Sound studio	understand how sound is sampled and stored in digital form	Pixel studio
15 Compression works	know why data is compressed	Sound studio
16 Revision HQ 1.2 part 2	know which parts of data representation I am secure on	Compression works
17 Assessment 2	assesses the whole unit	Revision HQ 1.2 part 2
18 Binary blitz	stretches what is taught	Assessment 2

The assessments in this unit are placed where it is so that the reflection sheet still has lessons left to send a pupil back to.

What pupils find hard here

None of this unit's walls names a common mistake in so many words. The misconceptions it works against are in the wrong options of its questions instead: every one of those is a plausible wrong idea rather than filler, and the lesson plans list them question by question.

Every multiple-choice and Predict activity in the unit — 164 of them — offers wrong answers that are the misconception, not filler. The split across those options at the hinge question is the cheapest diagnostic in the lesson.

Where the explaining, modelling and practising happen

PHASE	WHERE IT IS
Retrieve	The starter on the worksheet, before any device is opened. It is answered on paper and corrected before the lesson moves on.
Explain	The slides and the station walls carry the same explanation, so a teacher can teach from the front and a pupil can re-read it in the room.
Model	There is no worked example on screen in this unit: its activities are questions to answer rather than techniques to copy. The modelling is yours, from the slides, before anyone starts - and it is the part to protect when the lesson runs short.
Practise	166 guided activities, then 58 independent ones.
Check	The badge questions mark as the pupil goes; the hinge question is for the class at once.
Apply	0 challenge activities, and the exam practice on the worksheet.
Review	The exit question and the confidence columns. The confidence columns supplement the evidence; they do not stand in for it.

How the website, the slides and the worksheet divide the work

They are not three copies of the same thing.

- **The slides** are for the front of the room: the explanation and the model, before anyone is working alone.
- **The experience** is where a pupil reads, decides and is marked. It gives the immediate feedback a teacher cannot give thirty pupils at once.
- **The worksheet** is the record and the writing. The key fact in their own words, the station challenge, the exam practice. It is also what survives when the devices do not.

The rule that holds them together is **write first, then answer at the badge**. A pupil who taps first has turned the lesson into a quiz game, and the worksheet is how you can see at a glance who did.

Why a 360° room, and what it should produce

The room is not there to be impressive. It is there because this unit's content is a set of related parts that a pupil has to hold together, and putting one part on each wall makes the set visible and navigable: a pupil can turn back to station 2 while answering station 4. The evidence it should produce is on paper — 78 key facts and 90 station challenges in a pupil's own words — and in the score screen, which shows which station a pupil lost marks on rather than one number.

No claim is made here that immersion improves learning on its own. What the format does is make the structure of a topic physical and keep a pupil's attention on one part at a time; the teaching is in the explanation, the example and the practice, as it would be anywhere else.

Support, and how it is taken away

- The worked example is on the early activities of a station and not on the late ones. That fade is deliberate.
- The hint is a ladder asked for one rung at a time, and the top of it is still the technique on different data, never the answer.
- Read aloud is on every task.
- Prescribed values are boxed and in a monospace face as well as coloured, so they survive a black-and-white printout and a high-contrast mode.
- **Lower literacy and EAL:** pre-teach the vocabulary (RAM, ROM, Virtual memory, Volatile, Secondary storage, Non-volatile, Capacity, Durability, Portability, Suitability, Bit, Byte). Accept the key fact spoken before it is written, and in two or three words.
- **SEND:** pace is not fixed. Progress saves after every answer and reopens at the earliest unfinished question, so a pupil may finish a station next lesson without losing anything.

- A headset is never required.

Checking, and what to do with what you find

Three checks, in this order: the starter (did last lesson stick), the hinge question (is the class ready for independent work), the exit question (did this lesson stick). The first two change what you do in the lesson; the third changes what you start the next one with.

When a hinge question splits the class, go back to the station it sits on and work its example again, aloud, before anyone goes on. When a test comes back, the reflection sheet names the topics to revisit and review mode on those lessons reopens exactly the questions a pupil got wrong.

Where the approach comes from

The shape of a lesson here — retrieve, explain, model, practise with support that fades, check, apply, review — is ordinary evidence-informed practice rather than anything this product invented. Useful starting points, none of them about Revise 360:

- Rosenshine, *Principles of Instruction* (American Educator, 2012) — the daily review, small steps, worked examples and checking for understanding that this structure follows.
- Sweller, van Merriënboer and Paas on cognitive load and the worked-example effect — why the example comes before the independent task and why the support is withdrawn gradually.
- Dunlosky et al., *Improving Students' Learning With Effective Learning Techniques* (2013) — the evidence for retrieval practice and spacing, which the starters and the review mode use.
- Wiliam, *Embedded Formative Assessment* — hinge questions and what to do with the answers.

What is not claimed. Nothing above is evidence that Revise 360 improves outcomes. No trial of this product has been run, and immersion is not claimed to improve learning by itself. The citations are for the teaching structure; whether these resources work in your classroom is a question your own assessment answers.

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